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COMMON HALF YEARLY EXAMINATION – 2025

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Standard XII

Reg.No. | | | | | | |

CHEMISTRY

Time: 3.00 hrs. Part – I Marks: 70

I. Choose the correct answer: 15 x 1 = 15

1. Choose the correct one.
   I. Haematite - Sulphide ore
   II. Copper pyrite - Oxide ore
   III. Galena - Sulphate ore
   IV. Silver glance - Sulphide ore.
   a) I, II and III b) IV only c) I and IV d) III only
2. Which of the following statements is not correct?
   a) Beryl is a cyclic silicate b) Mg_2SiO_4 is an orthosilicate.
   c) SiO_4^{4-} is the basic structural unit of silicates
   d) Feldspar is not aluminosilicate
3. The correct order of the thermal stability of hydrogen halide is
   a) HI > HBr > HCl > HF b) HF > HCl > HBr > HI
   c) HCl > HF> HBr > HI d) HI > HCl > HF> HBr
4. Among the transition metals of 3d series, the one that has highest negative (M^{2+}/M) standard electrode potential is
   a) Ti b) Cu c) Mn d) Zn
5. Ni^{2+} ions present in nickel chloride solution is estimated accurately using the ligand.
   a) DMG b) EDTA c) en d) CO
6. The fraction of total volume occupied by the atoms in a simple cubic is
   a) $\frac{\pi}{4\sqrt{2}}$ b) $\frac{\pi}{6}$ c) $\frac{\pi}{4}$ d) $\frac{\pi}{3\sqrt{2}}$
7. Assertion (A) : Rate of a reaction can be written as $\frac{-d\,[Reactant]}{dt}$
   Reason (R) : As time increases concentration of reactant decrease.
   a) Both A and R are correct and R explains A b) A is correct but R is wrong
   c) A is wrong but R is correct
   d) Both A and R are correct but R does not explain A
8. Which of the following fluro compounds is most likely to behave as a Lewis base?
   a) BF_3 b) PF_3 c) CF_4 d) SiF_4
9. How many Faradays of electricity are required for the following reaction to occur?
   $MnO^-_4 \rightarrow Mn^{2+}$ a) 5F b) 3F c) 1F d) 7F
10. The movement of dispersed phase under the influence of electric current is known as
    a) Electro Osmosis b) Electrophoresis
    c) Electro dialysis d) Ultra filteration
11. Williamson synthesis of preparing dimethyl ether is
    a) SN^1 reactions b) SN^2 reactions
    c) electrophilic addition d) electrophilic substitution
12. Which one of the following undergoes reaction with 50% sodium hydroxide solution to give the corresponding alcohol and acid?
    a) phenyl methanal b) ethanal c) ethanol d) methanol.
13. The product formed by the reaction an aldehyde with a primary amine
    a) Carboxylic acid b) Aromatic acid c) Schiff's base d) Ketone
14. Gun cotton is a) cellulose acetate
    b) cellulose bromide c) ethyl cellulose d) cellulose nitrate
15. The polymer used in making blankets (artificial wool) is
    a) Polystyrene b) PAN c) Polyester d) Polythene

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Part - II

II. Answer any 6 questions. (Q.No.24 is compulsory) **6 x 2 = 12**

16. What is Auto reduction?
17. Write the conditions given by Hume-Rothery's rule for the formation of alloy.
18. Write the IUPAC name of co-ordination compounds.
    a) $[Co(NH_3)_3(Cl)_3]$ b) $[Ag(CN)_2]^-$
19. Distinguish between Isotropy and Anisotropy in solids.
20. Why is AC current used instead of DC in measuring the electrolytic conductance?
21. What are promoters? Give example.
22. What are hormones? Give example.
23. Draw the structure of a) Aspirin b) Paracetamol
24. Find the compound A and B.

$$CH_3 - CH_2 - Cl \xrightarrow{\text{alc.KOH}} A \xrightarrow[298\ K]{H_2SO_4/H_2O} B$$

Part - III

III. Answer any 6 questions. (Q.No.33 is compulsory) **6 x 3 = 18**

25. Write the uses of Potash alum.
26. Explain the structure of NH_3.
27. Give any three oxidation reaction of $K_2Cr_2O_7$
28. What is pseudo first order reaction? Give example.
29. Explain common ion effect with example.
30. Short note on "Gold number".
31. How is Picric acid prepared?
32. Explain the reducing nature of formic acid.
33. Identify compounds A, B and C.

$$C_6H_5{-}NO_2 \xrightarrow{Fe/HCl} A \xrightarrow[273\ K]{HNO_2} B \xrightarrow{C_6H_5OH} C$$

Part - IV

IV. Answer all the questions. **5 x 5 = 25**

34. a) i) Explain the Van-Arkel method. (3m)
    ii) CO is a reducing agent. Justify with an example. (2m) ***(OR)***
    b) i) How do you prepare Bleaching powder. (2m)
    ii) Compare Lanthanides and Actinides. (3m)
35. a) Explain the Werner's theory. (5m) ***(OR)***
    b) Write the general characteristics of solids. (5m)
36. a) Derive Henderson-Hasselbalch equations (5m). ***(OR)***
    b) i) After 120 minutes a radioactive substance becomes $\left(\frac{1}{16}\right)^{th}$ of original amount. Find the half life period. (2m)
    ii) Write a note on SHE. (3m)
37. a) Write note on
    i) Riemer–Tiemann reaction (3m) ii) Carbylamine reaction (2m) ***(OR)***
    b) Explain the structure of glucose. (5m) .
38. a) Explain the mechanism of cleaning action of soaps and detergents. (5m) ***(OR)***
    b) An organic compound (A) C_2H_4O reduce Tollen's and Fehling's solution. (A) react with methanol and HCl to give compound (B) $C_4H_{10}O_2$. Compound (A) on reaction with methanal in the presence of NaOH to give compound (C) $C_3H_6O_2$. Identify compounds A, B and C. with necessary reactions.

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